

MIT Integration Bee Qualifying Exam 2025 Problem 4

Problem:

$$\int_1^{e^e} \frac{\log(x^{x \log x})}{x^2} dx$$

Solution:

Solution. Using the standard convention $\log x = \ln x$, we simplify the integrand:

$$\log(x^{x \log x}) = x \log x \cdot \log x = x(\log x)^2.$$

Therefore,

$$\frac{\log(x^{x \log x})}{x^2} = \frac{x(\log x)^2}{x^2} = \frac{(\log x)^2}{x}.$$

So the integral becomes

$$\int_1^{e^e} \frac{(\log x)^2}{x} dx.$$

Let

$$u = \log x.$$

Then

$$du = \frac{1}{x} dx.$$

Changing the bounds:

$$x = 1 \implies u = \log 1 = 0,$$

and

$$x = e^e \implies u = \log(e^e) = e.$$

Thus,

$$\int_1^{e^e} \frac{(\log x)^2}{x} dx = \int_0^e u^2 du.$$

Now,

$$\int_0^e u^2 du = \left[\frac{u^3}{3} \right]_0^e = \frac{e^3}{3}.$$

Therefore,

$$\boxed{\frac{e^3}{3}}.$$

□